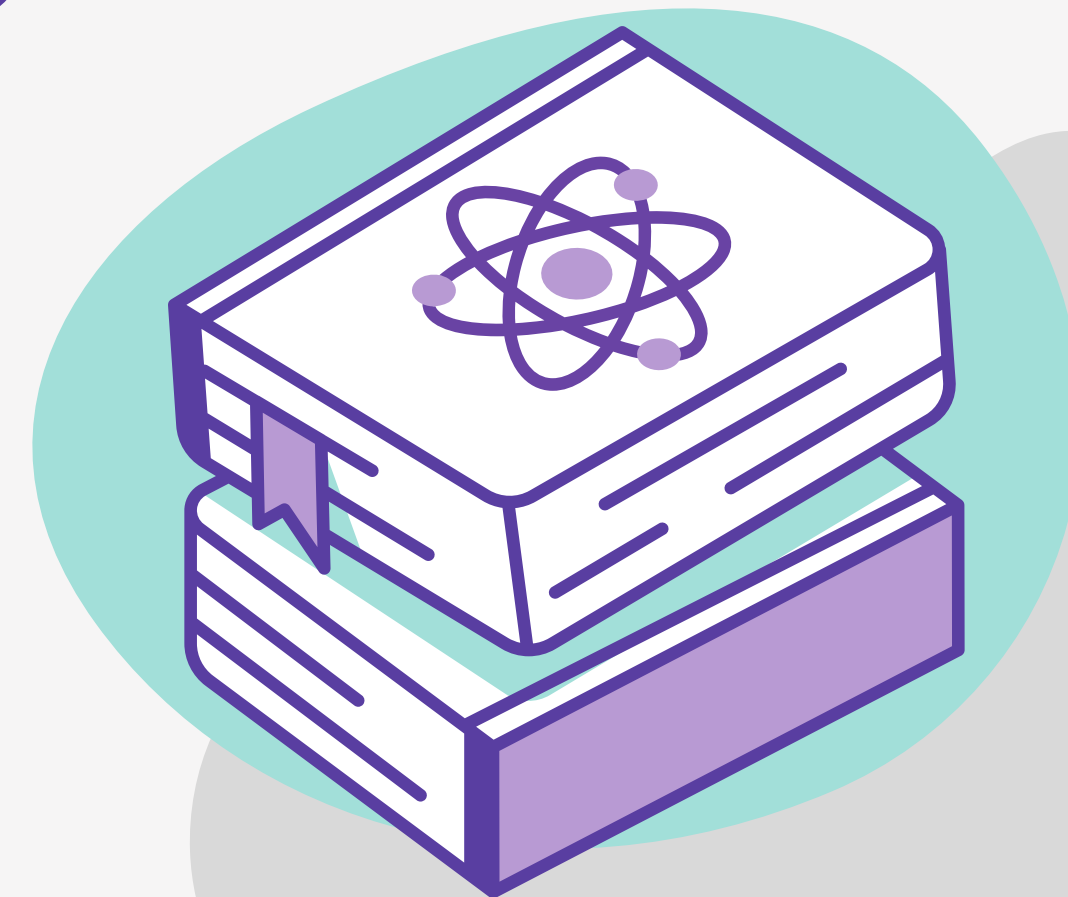


Term Project

**CHE251 : Chemical Process and
Calculations**

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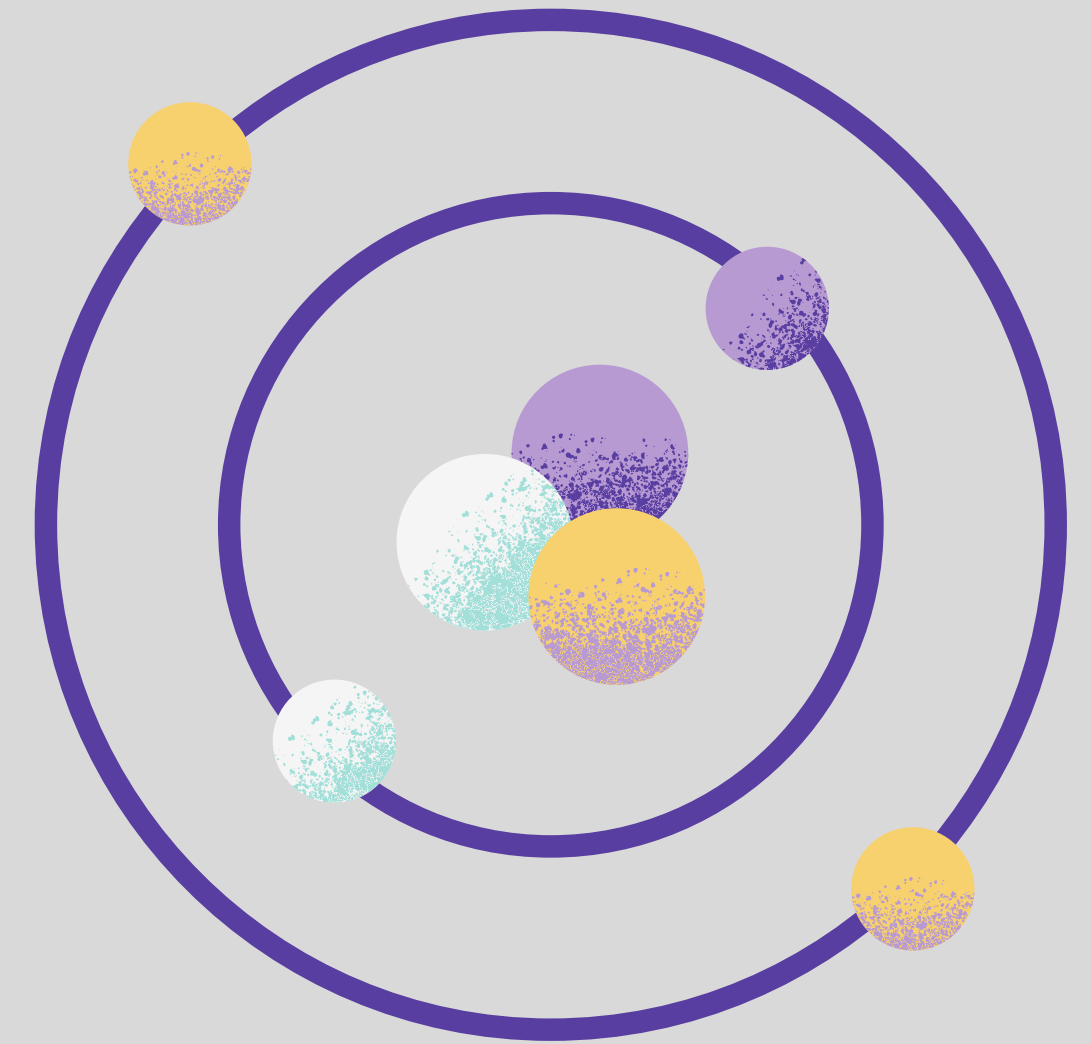


Abstract

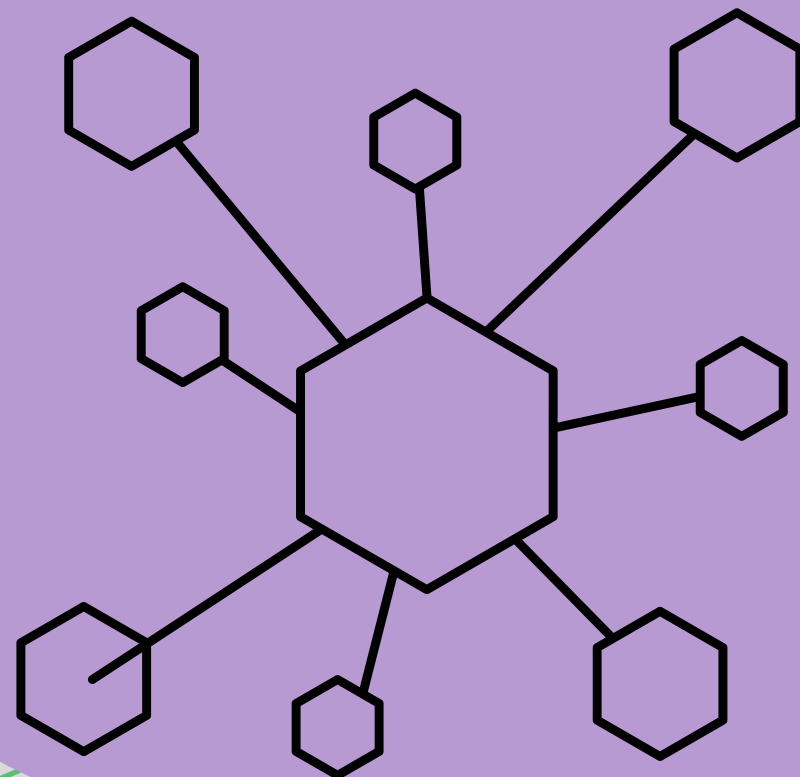
The project focuses on detailed material balance analysis for the Haber process, which synthesizes ammonia (NH_3) from nitrogen and hydrogen. Due to limited single-pass conversion, the system requires recycle and purge streams to achieve higher overall yields. The analysis evaluates how reactor conversion and purge ratio affect ammonia production, providing both theoretical insight and process optimization strategies.

Background & Motivation

- Ammonia is a critical chemical used in fertilizers and explosives.
- The Haber process combines N_2 and H_2 under high temperature (400–500°C) and pressure (150–250 atm).
- Single-pass conversion is only 20–30% due to equilibrium limitations.
- To improve yield, unreacted gases are separated, recycled, and partially purged.



Objectives



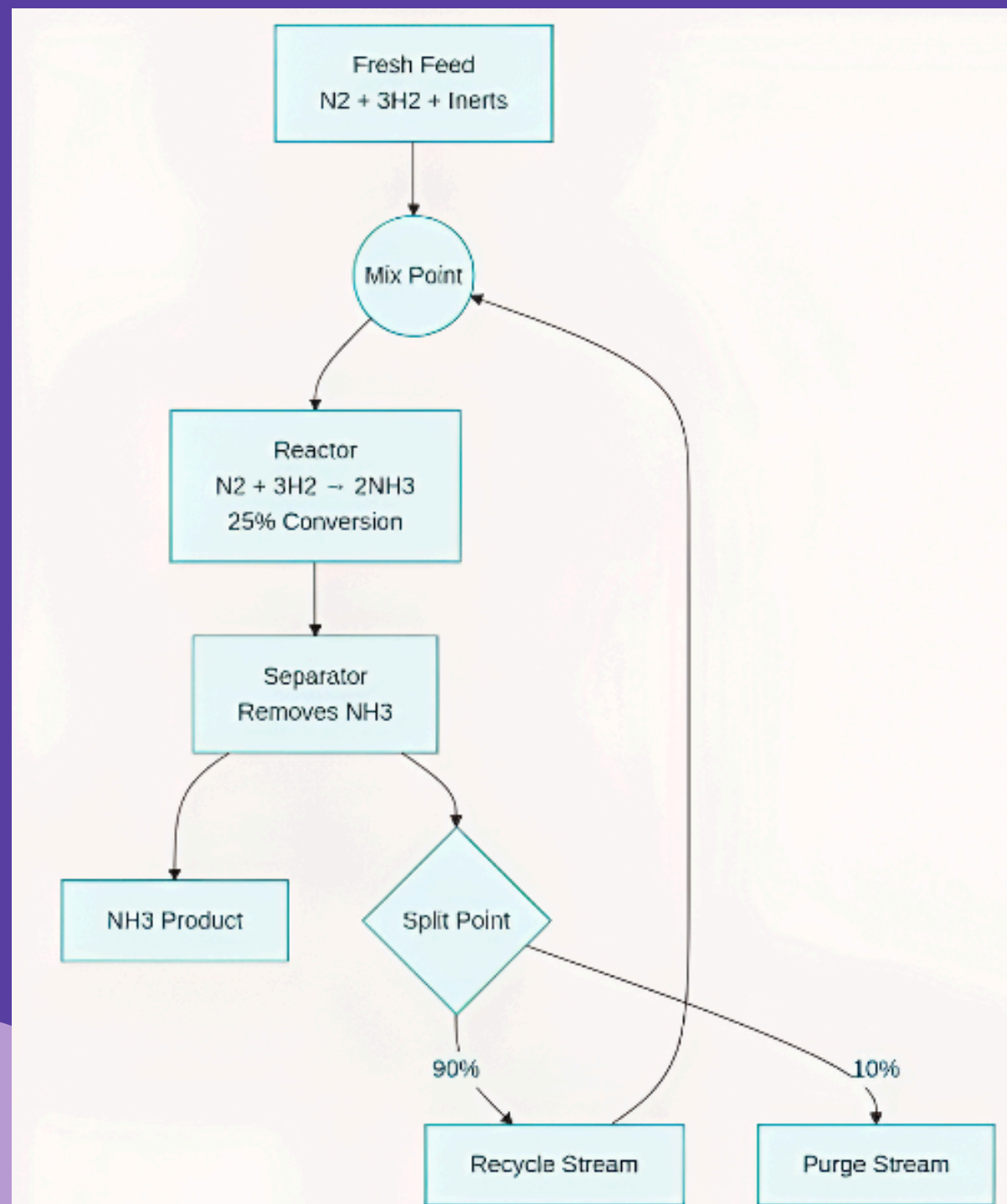
- . Develop a process flow diagram including reactor, separator, recycle, and purge units.
- . Derive material balance equations for each unit.
- . Quantify the impact of single-pass conversion and purge ratio on ammonia yield.
- . Establish steady-state relations for inert gas control.
- . Compare performance across different operating conditions.

Process Overview

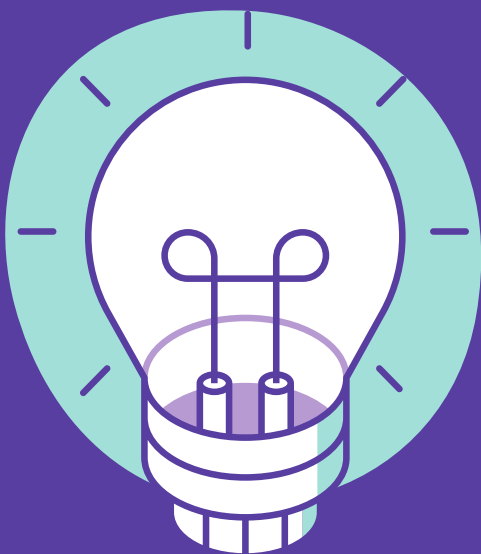


- The Haber process converts nitrogen and hydrogen to ammonia.
- $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ (exothermic)
- Main components:
- Fresh Feed – mixture of N_2 , H_2 , and small fraction of inerts.
- Reactor – partial conversion of N_2 and H_2 to NH_3

Process Flow Diagram

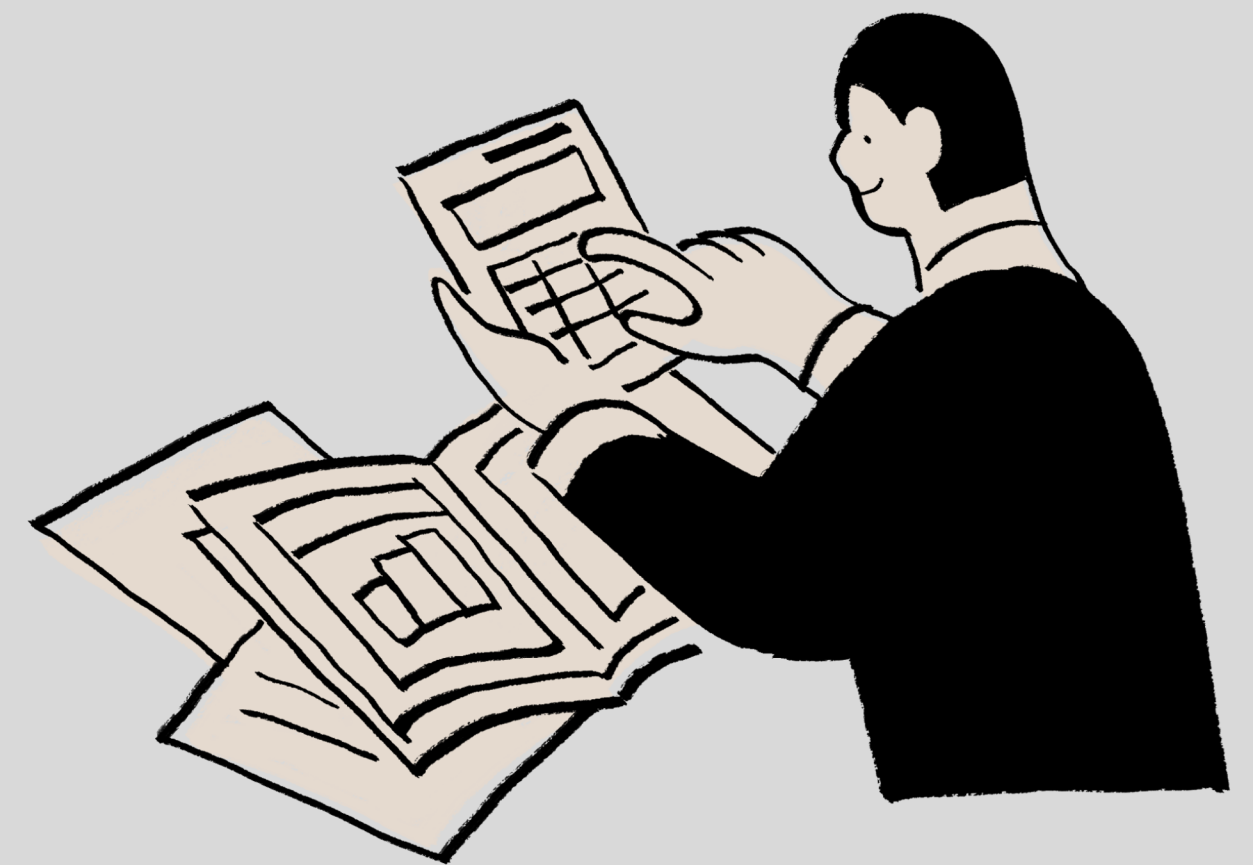


- Fresh feed enters the system containing N₂, H₂, and small inert fraction.
- In the reactor, partial conversion takes place.
- Separator removes produced NH₃.
- Unreacted gases form a mixed stream.
- This stream is split — one part recycled, another purged.
- Steady-state operation ensures constant inert concentration.



Assumptions and Basis

1. **Basis:** 100 mol fresh feed.
2. **Feed composition:** 24 mol N₂, 72 mol H₂, 4 mol inerts (this is just for example N₂ and H₂ to be taken in 1 : 3 ratio)
3. **Reaction:** $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$.
4. **Single-pass conversion :** x
5. **Purge ratio :** p
6. **Ideal separation** – all NH₃ removed as product.
7. **Steady state:** No accumulation in system.
8. **Reactor operates under equilibrium conditions.**



Material Balance

Notation (single, consistent set)

- Streams (mol):
 - F_{N_2} , F_{H_2} , F_I = fresh feed moles of N_2 , H_2 , and inerts.
 - N_{in} , H_{in} , I_{in} = mole in reactor feed (i.e., fresh + recycle entering reactor).
 - N_{out} , H_{out} , I_{out} = mole leaving reactor (before separator).
 - R_N , R_H , R_I = moles in the recycle stream returned to the reactor (after split).
 - P_N , P_H , P_I = moles in the purge stream (the portion removed to prevent inert buildup).
 - P_{NH_3} = product ammonia (all NH_3 removed in separator, per assumption).
- Parameters:
 - x = single-pass conversion of N_2 in the reactor feed (fraction of N_2 in reactor feed that reacts). Important: conversion is defined on reactor feed, not necessarily on fresh feed.
 - p = purge fraction (the fraction of the separator outlet that is purged). Equivalently the recycle split is $1-p$ (fraction returned).
- Stoichiometry: $N_2 + 3 H_2 \rightarrow 2 NH_3$.



Material Balance (Essential Concepts used and the Final Equations)

$x = (\text{moles entering reactor} - \text{moles leaving reactor}) / \text{moles entering reactor}$
 $x = (N_{in} - N_{out}) / N_{in} = N_{react} / N_{in}$

Outlet (before separator):

$\rightarrow N_{out} = N_{in} - N_{react} = (1-x)N_{in}$, $H_{out} = H_{in} - 3xN_{in}$, $I_{out} = I_{in}$ (inerts do not react)

Recycle & purge relations:

$\rightarrow R_N = (1-p)N_{out} = (1-p)(1-x)N_{in}$, $P_N = p \cdot N_{out} = p(1-x)N_{in}$, $R_H = (1-p)(H_{in} - 3xN_{in})$,
 $R_I = (1-p)I_{in}$

Reactor feed = fresh + recycle

$\rightarrow N_{in} = F_N + R_N \Rightarrow N_{in} = F_N / (1 - (1-p)(1-x))$

Product(NH₃)

$\rightarrow P_{NH_3} = 2x \cdot N_{in} = 2x \cdot F_N / (1 - (1-p)(1-x))$



Calculation Methodology

- Use overall and component balances to establish equations.
- Solve for unreacted gas compositions using assumed x and p .
- Determine NH_3 yield for different values of x and p .
- Analyze results to identify optimal operational conditions.

Tools used:

- MATLAB for solving nonlinear balance equations.
- x - p trade-off analysis.

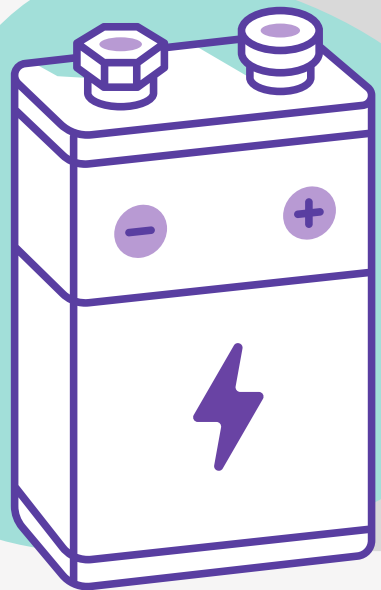
Results

Case Studies:

1. $x = 25\%$, $p = 10\% \rightarrow \text{NH}_3 \text{ yield} = 14.8 \text{ mol}/100 \text{ mol feed}.$
2. $x = 30\%$, $p = 5\% \rightarrow \text{NH}_3 \text{ yield} = 21.5 \text{ mol}/100 \text{ mol feed}.$
3. $x = 20\%$, $p = 15\% \rightarrow \text{NH}_3 \text{ yield} = 10.5 \text{ mol}/100 \text{ mol feed}.$

Observations:

1. Yield increases with x .
2. Yield decreases with p .
3. There exists an optimal purge range ($\sim 3\text{--}5\%$) for stable operation



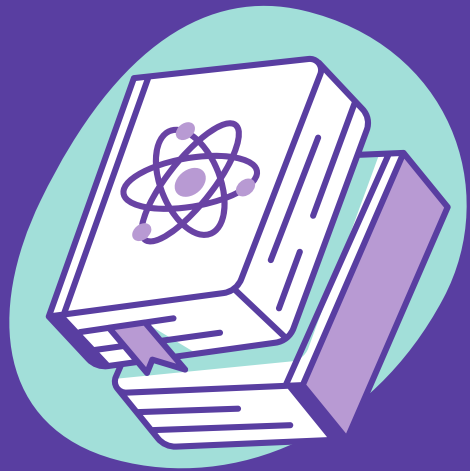
Discussion Insights

- Increasing reactor conversion directly enhances yield.
- Higher purge reduces accumulation but wastes reactants.
- Zero purge leads to inert buildup, reducing efficiency.
- Process optimization must balance purge and recycle.
- Industrial processes typically operate at $x = 25\text{--}30\%$ and $p = 3\text{--}5\%$.



Conclusion

- The steady-state material balance for ammonia production reveals that:
- Recycle significantly increases yield beyond single-pass limits.
- Purge is essential for inert control.
- Optimized conversion and purge ratios ensure efficient operation.
- The developed framework can be extended for sensitivity analysis and validation with industrial plant data.



References



1. Felder & Rousseau, Elementary Principles of Chemical Processes, Wiley.
2. Himmelblau, Basic Principles and Calculations in Chemical Engineering.
3. Smith, Van Ness & Abbott, Introduction to Chemical Engineering Thermodynamics.

THANK YOU